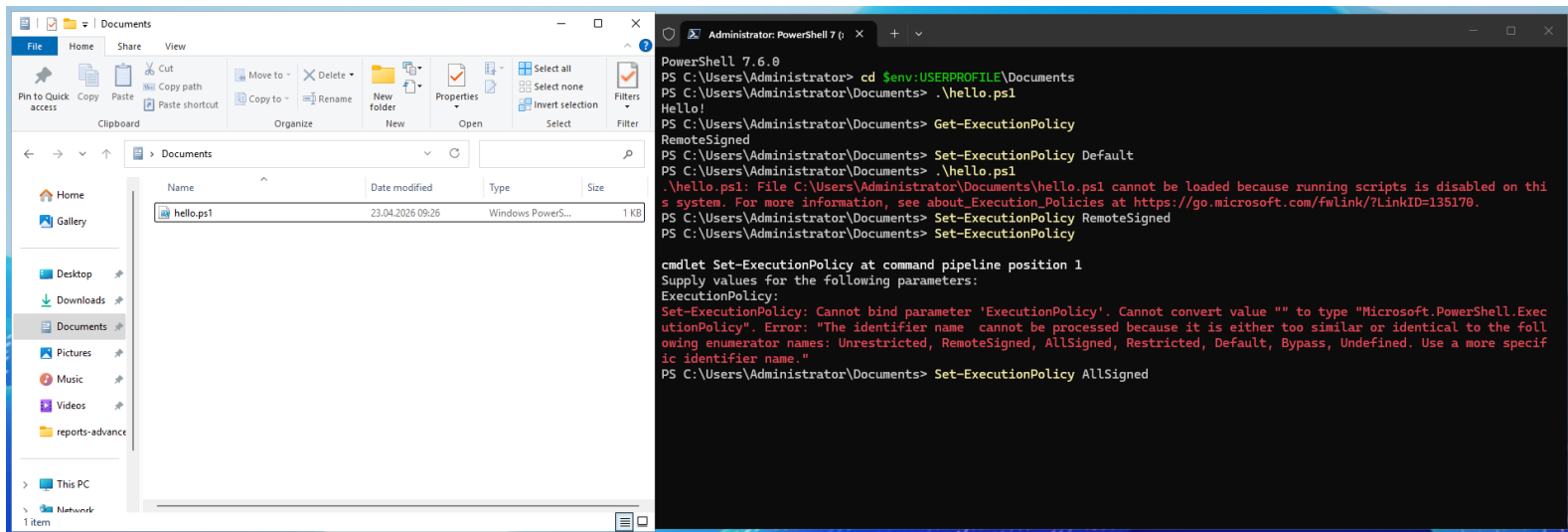
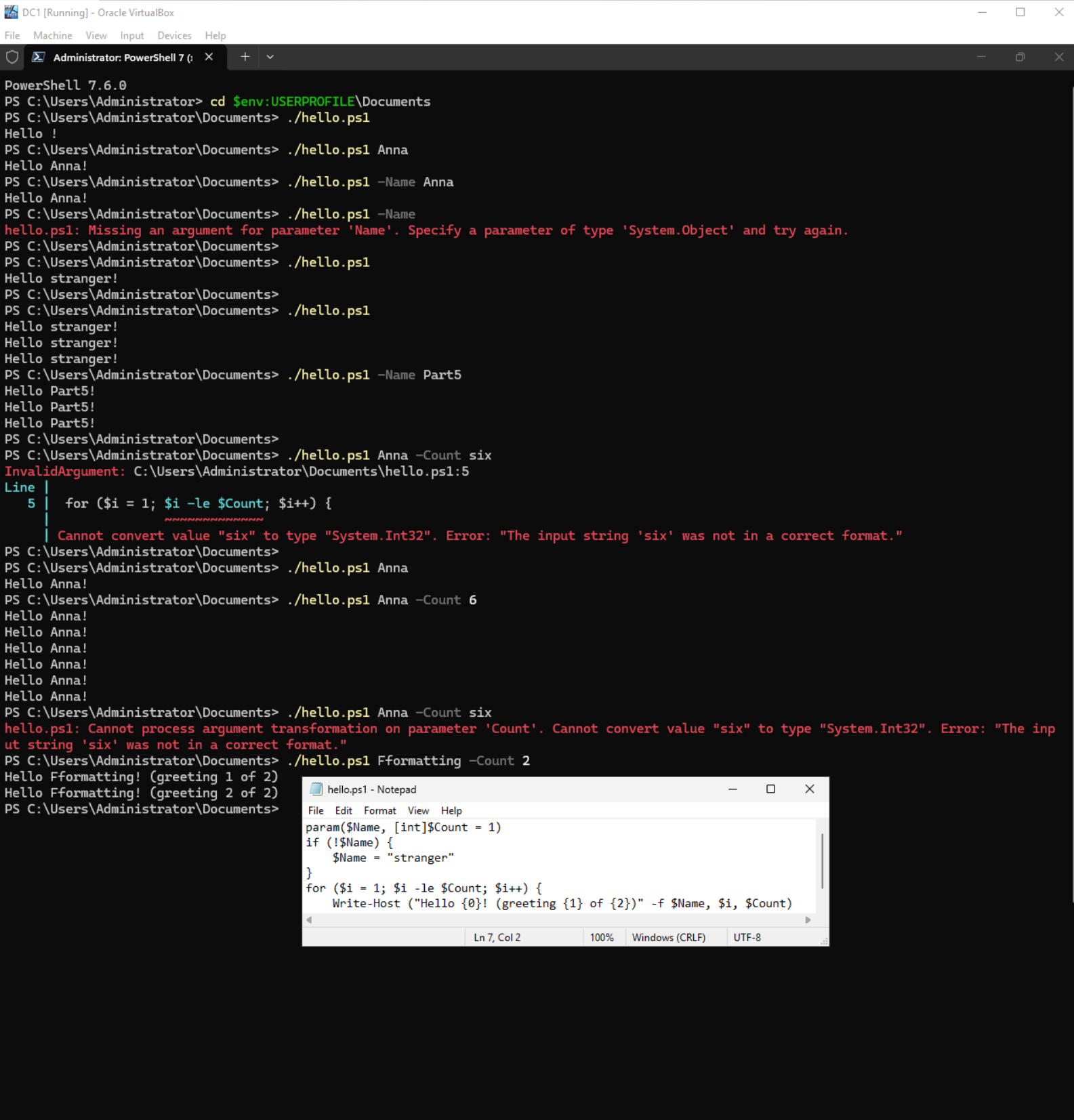


# Lab 8 – My Solutions

## 3.1: Running a Script, Execution Policy:



### 3.2: Using Variables, Loops, and Conditional Statements:



```
PowerShell 7.6.0
PS C:\Users\Administrator> cd $env:USERPROFILE\Documents
PS C:\Users\Administrator\Documents> ./hello.ps1
Hello !
PS C:\Users\Administrator\Documents> ./hello.ps1 Anna
Hello Anna!
PS C:\Users\Administrator\Documents> ./hello.ps1 -Name Anna
Hello Anna!
PS C:\Users\Administrator\Documents> ./hello.ps1 -Name
hello.ps1: Missing an argument for parameter 'Name'. Specify a parameter of type 'System.Object' and try again.
PS C:\Users\Administrator\Documents> ./hello.ps1
Hello stranger!
PS C:\Users\Administrator\Documents> ./hello.ps1
Hello stranger!
PS C:\Users\Administrator\Documents> ./hello.ps1
Hello stranger!
PS C:\Users\Administrator\Documents> ./hello.ps1 -Name Part5
Hello Part5!
Hello Part5!
Hello Part5!
PS C:\Users\Administrator\Documents>
PS C:\Users\Administrator\Documents> ./hello.ps1 Anna -Count six
InvalidArgument: C:\Users\Administrator\Documents\hello.ps1:5
Line |
5 | for ($i = 1; $i -le $Count; $i++) {
      ~~~~~
      Cannot convert value "six" to type "System.Int32". Error: "The input string 'six' was not in a correct format."
PS C:\Users\Administrator\Documents>
PS C:\Users\Administrator\Documents> ./hello.ps1 Anna
Hello Anna!
PS C:\Users\Administrator\Documents> ./hello.ps1 Anna -Count 6
Hello Anna!
Hello Anna!
Hello Anna!
Hello Anna!
Hello Anna!
PS C:\Users\Administrator\Documents> ./hello.ps1 Anna -Count six
hello.ps1: Cannot process argument transformation on parameter 'Count'. Cannot convert value "six" to type "System.Int32". Error: "The input string 'six' was not in a correct format."
PS C:\Users\Administrator\Documents> ./hello.ps1 Fformatting -Count 2
Hello Fformatting! (greeting 1 of 2)
Hello Fformatting! (greeting 2 of 2)
PS C:\Users\Administrator\Documents>
```

```
hello.ps1 - Notepad
File Edit Format View Help
param($Name, [int]$Count = 1)
if (!$Name) {
    $Name = "stranger"
}
for ($i = 1; $i -le $Count; $i++) {
    Write-Host ("Hello {0}! (greeting {1} of {2})" -f $Name, $i, $Count)
}
```

### 3.3: Communication with the System

The image shows two Oracle VM VirtualBox windows. The top window, titled 'CL1 [Running] - Oracle VirtualBox', contains a PowerShell terminal window titled 'Administrator: PowerShell 7 (x)'. The terminal displays a list of system processes with columns for PID, Private Bytes (PM), Working Set (WS), CPU time, ID, Session ID (SI), Process Name, and PSComputerName. The processes listed include svchost, System, taskhostw, VBoxService, VBoxTray, vds, WindowsTerminal, wininit, winlogon, wLms, wsmprovhost, and wsmprovhost. Below the list, a PowerShell command is executed: `Invoke-Command -ComputerName DC1 -ScriptBlock { Get-Process | Where-Object name -eq notepad }`. The output shows a single process: notepad (PID 1152, PM 2.94, WS 23.72, CPU 0.09, ID 1). Another command is executed: `Invoke-Command -ComputerName DC1 -ScriptBlock { Stop-Process -Name notepad }`. The bottom window, titled 'DC1 [Running] - Oracle VirtualBox', contains a PowerShell terminal window titled 'Administrator: PowerShell 7 (x)'. The terminal displays the output of `$notes.CloseMainWindow()`, which returns `True`. The terminal also shows the output of `$notes = (Get-Process | Where name -eq notepad)`, which returns an empty array. The terminal then displays the output of `$notes.CloseMainWindow()`, which returns `True`. The terminal also shows the output of `Update-Help`, which returns `PS C:\Users\Administrator> |`.

CL1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Administrator: PowerShell 7 (x)

| PID | PM    | WS    | CPU   | ID   | SI | ProcessName     | PSComputerName |
|-----|-------|-------|-------|------|----|-----------------|----------------|
| 17  | 5.18  | 24.56 | 7.34  | 6228 | 0  | svchost         | DC1            |
| 12  | 2.66  | 14.88 | 0.16  | 6592 | 1  | svchost         | DC1            |
| 13  | 2.56  | 14.75 | 0.05  | 6832 | 0  | svchost         | DC1            |
| 18  | 12.67 | 32.79 | 0.69  | 7140 | 0  | svchost         | DC1            |
| 16  | 3.74  | 25.13 | 0.12  | 7240 | 0  | svchost         | DC1            |
| 13  | 3.22  | 19.58 | 0.45  | 7256 | 1  | svchost         | DC1            |
| 13  | 2.74  | 15.13 | 0.11  | 7828 | 1  | svchost         | DC1            |
| 0   | 0.04  | 0.17  | 35.28 | 4    | 0  | System          | DC1            |
| 25  | 5.14  | 17.12 | 0.16  | 3324 | 1  | taskhostw       | DC1            |
| 11  | 2.16  | 8.00  | 1.03  | 1772 | 0  | VBoxService     | DC1            |
| 12  | 2.74  | 13.39 | 0.44  | 2244 | 1  | VBoxTray        | DC1            |
| 16  | 2.30  | 12.62 | 0.16  | 4748 | 0  | vds             | DC1            |
| 32  | 31.98 | 93.65 | 90.89 | 5088 | 1  | WindowsTerminal | DC1            |
| 12  | 1.62  | 10.16 | 0.06  | 668  | 0  | wininit         | DC1            |
| 12  | 2.45  | 13.38 | 0.38  | 740  | 1  | winlogon        | DC1            |
| 6   | 0.68  | 4.44  | 0.02  | 1816 | 0  | wLms            | DC1            |
| 29  | 43.79 | 63.61 | 2.84  | 6092 | 0  | wsmprovhost     | DC1            |
| 30  | 59.30 | 81.82 | 2.81  | 7120 | 0  | wsmprovhost     | DC1            |

PS C:\Users\Administrator.CONTOSO> Invoke-Command -ComputerName DC1 -ScriptBlock { Get-Process | Where-Object name -eq notepad }

| NPM(K) | PM(M) | WS(M) | CPU(s) | Id   | SI | ProcessName | PSComputerName |
|--------|-------|-------|--------|------|----|-------------|----------------|
| 14     | 2.94  | 23.72 | 0.09   | 1152 | 1  | notepad     | DC1            |

PS C:\Users\Administrator.CONTOSO> Invoke-Command -ComputerName DC1 -ScriptBlock { Stop-Process -Name notepad }

PS C:\Users\Administrator.CONTOSO>

DC1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Administrator: PowerShell 7 (x)

Threads Property System.Diagnostics.ProcessThreadCollection Threads {get;}

TotalProcessorTime Property timespan TotalProcessorTime {get;}

UserProcessorTime Property timespan UserProcessorTime {get;}

VirtualMemorySize Property int VirtualMemorySize {get;}

VirtualMemorySize64 Property long VirtualMemorySize64 {get;}

WorkingSet Property int WorkingSet {get;}

WorkingSet64 Property long WorkingSet64 {get;}

PSConfiguration PropertySet PSConfiguration {Name, Id, PriorityClass, FileVersion}

PSResources PropertySet PSResources {Name, Id, Handlecount, WorkingSet, NonPagedMemorySize, PagedMemorySize, PrivateMe...

CommandLine ScriptProperty System.Object CommandLine {get=...

Company ScriptProperty System.Object Company {get=\$this.Mainmodule.FileVersionInfo.CompanyName;}

CPU ScriptProperty System.Object CPU {get=\$this.TotalProcessorTime.TotalSeconds;}

Description ScriptProperty System.Object Description {get=\$this.Mainmodule.FileVersionInfo.FileDescription;}

FileVersion ScriptProperty System.Object FileVersion {get=\$this.Mainmodule.FileVersionInfo.FileVersion;}

Path ScriptProperty System.Object Path {get=\$this.Mainmodule.FileName;}

Product ScriptProperty System.Object Product {get=\$this.Mainmodule.FileVersionInfo.ProductName;}

ProductVersion ScriptProperty System.Object ProductVersion {get=\$this.Mainmodule.FileVersionInfo.ProductVersion;}

PS C:\Users\Administrator> \$notes.CloseMainWindow()

True

PS C:\Users\Administrator> \$notes = (Get-Process | Where name -eq notepad)

PS C:\Users\Administrator> \$notes.CloseMainWindow()

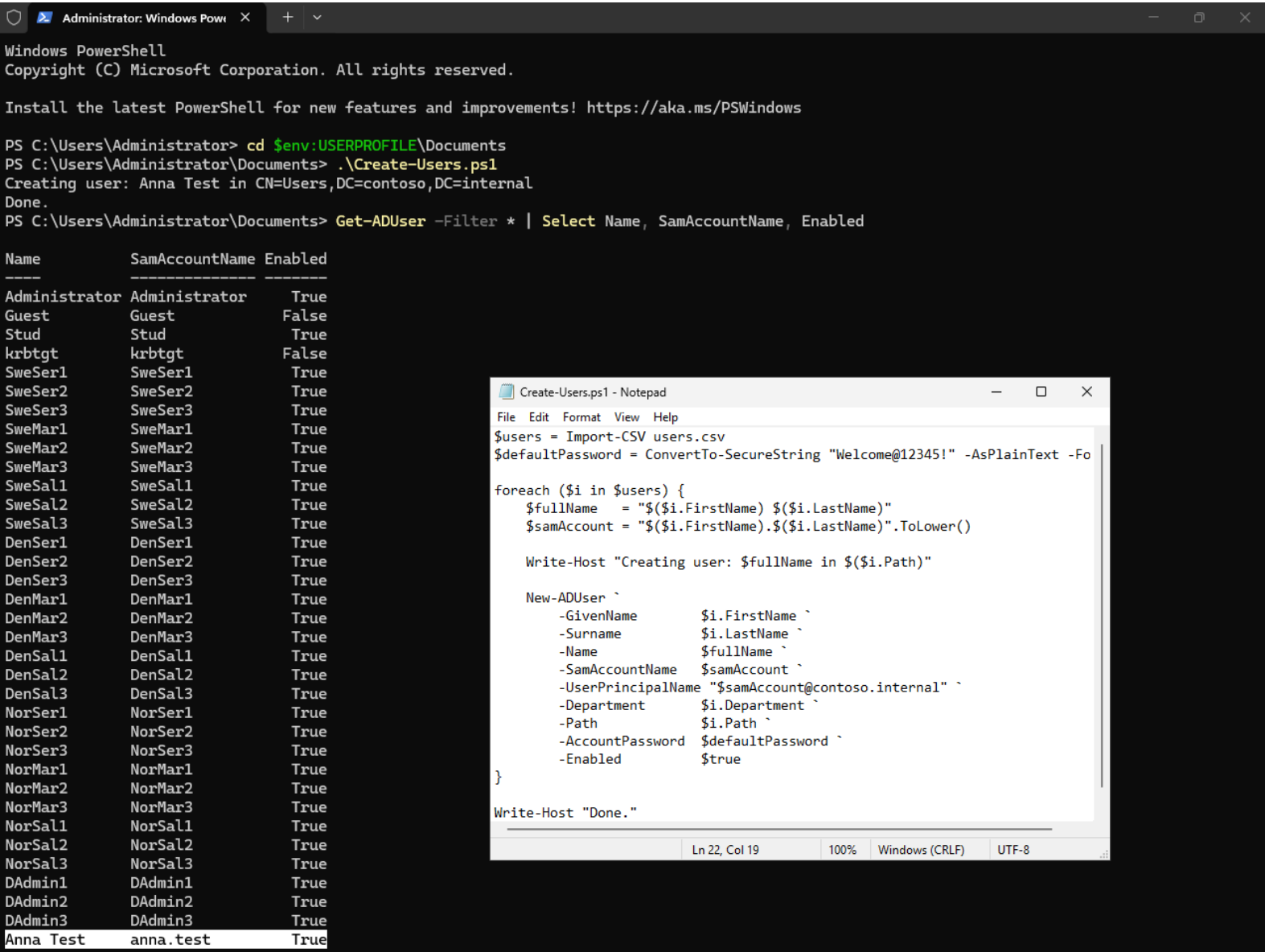
True

True

PS C:\Users\Administrator> Update-Help

PS C:\Users\Administrator> |

### 3.4: Sharing Folders via PowerShell



The screenshot shows a Windows PowerShell terminal window with the following commands and output:

```
PS C:\Users\Administrator> cd $env:USERPROFILE\Documents
PS C:\Users\Administrator\Documents> .\Create-Users.ps1
Creating user: Anna Test in CN=Users,DC=contoso,DC=internal
Done.
PS C:\Users\Administrator\Documents> Get-ADUser -Filter * | Select Name, SamAccountName, Enabled
```

| Name          | SamAccountName | Enabled |
|---------------|----------------|---------|
| Administrator | Administrator  | True    |
| Guest         | Guest          | False   |
| Stud          | Stud           | True    |
| krbtgt        | krbtgt         | False   |
| SweSer1       | SweSer1        | True    |
| SweSer2       | SweSer2        | True    |
| SweSer3       | SweSer3        | True    |
| SweMar1       | SweMar1        | True    |
| SweMar2       | SweMar2        | True    |
| SweMar3       | SweMar3        | True    |
| SweSal1       | SweSal1        | True    |
| SweSal2       | SweSal2        | True    |
| SweSal3       | SweSal3        | True    |
| DenSer1       | DenSer1        | True    |
| DenSer2       | DenSer2        | True    |
| DenSer3       | DenSer3        | True    |
| DenMar1       | DenMar1        | True    |
| DenMar2       | DenMar2        | True    |
| DenMar3       | DenMar3        | True    |
| DenSal1       | DenSal1        | True    |
| DenSal2       | DenSal2        | True    |
| DenSal3       | DenSal3        | True    |
| NorSer1       | NorSer1        | True    |
| NorSer2       | NorSer2        | True    |
| NorSer3       | NorSer3        | True    |
| NorMar1       | NorMar1        | True    |
| NorMar2       | NorMar2        | True    |
| NorMar3       | NorMar3        | True    |
| NorSal1       | NorSal1        | True    |
| NorSal2       | NorSal2        | True    |
| NorSal3       | NorSal3        | True    |
| DAdmin1       | DAdmin1        | True    |
| DAdmin2       | DAdmin2        | True    |
| DAdmin3       | DAdmin3        | True    |
| Anna Test     | anna.test      | True    |

The Notepad window shows the content of the `Create-Users.ps1` script:

```
File Edit Format View Help
$users = Import-CSV users.csv
$defaultPassword = ConvertTo-SecureString "Welcome@12345!" -AsPlainText -Fo

foreach ($i in $users) {
    $fullName = "$($i.FirstName) $($i.LastName)"
    $samAccount = "$($i.FirstName).$($i.LastName)".ToLower()

    Write-Host "Creating user: $fullName in $($i.Path)"

    New-ADUser `
        -GivenName      $i.FirstName `
        -Surname         $i.LastName `
        -Name            $fullName `
        -SamAccountName  $samAccount `
        -UserPrincipalName "$samAccount@contoso.internal" `
        -Department     $i.Department `
        -Path            $i.Path `
        -AccountPassword $defaultPassword `
        -Enabled         $true
}

Write-Host "Done."
```